

Unseen Seminar Questions 5: Answers

Probability and Statistical Inference

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1. These three sampling schemes illustrate different conditions behind a law of large numbers. For each scheme, describe the behavior of $\bar{X}_n$ and identify the condition that holds or fails.

- (a) The X_i are iid Poisson(4), with finite mean 4.
- (b) $X_i = 4 + V$ for every i , where V is one nonconstant, mean-zero random variable shared by all observations.
- (c) The X_i are independent and $X_i \sim N(4, 2i)$, where $2i$ is the variance.

Solution.

- (a) The iid variables have finite mean 4, so the usual law of large numbers gives $\bar{X}_n \xrightarrow{p} 4$. The observations are independent, and they all have the same distribution.
- (b) Here $\bar{X}_n = 4 + V$ for every n . It therefore retains the same distribution and does not converge in probability to 4. Every observation has the same marginal distribution, but perfect dependence means that adding observations provides no new information.
- (c) The mean remains 4, but

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n 2i = \frac{n(n+1)}{n^2} = \frac{n+1}{n} \rightarrow 1.$$

Since the average is normal, $\bar{X}_n \xrightarrow{d} N(4, 1)$ rather than concentrating at 4. Independence holds, but the observations are not identically distributed and their variances grow with i . As a result, the variance of the average does not tend to zero.

2. Put the following OLS-consistency statements in a logical order and use them to complete the proof:

- A. $\hat{\beta} - \beta = \left(\frac{1}{n} \sum x_i x_i' \right)^{-1} \left(\frac{1}{n} \sum x_i \varepsilon_i \right),$
- B. $n^{-1} \sum x_i \varepsilon_i \xrightarrow{p} \mathbb{E}(x_i \varepsilon_i) = 0,$
- C. $n^{-1} \sum x_i x_i' \xrightarrow{p} Q, \quad Q \text{ nonsingular},$
- D. $\hat{\beta} - \beta \xrightarrow{p} Q^{-1} 0 = 0.$

Which line uses A3', $\mathbb{E}(x_i \varepsilon_i) = 0$? Which line uses sufficient regressor variation? What would the probability limit be if the limit in B were $g \neq 0$? You may use the probability-limit rules for products and inverses used in the lecture's OLS-consistency argument; you do not need to state or prove the named theorems behind these rules.

Solution. Start with the algebraic decomposition A. Next apply the two laws of large numbers B and C; their relative order does not matter. Finally, use the permitted probability-limit rules for the inverse and product to reach D. Line B uses A3', the zero regressor-error moment $\mathbb{E}(x_i \varepsilon_i) = 0$.

Line C, especially the nonsingularity of Q , uses sufficient regressor variation. If the numerator converged to $g \neq 0$, then

$$\hat{\beta} - \beta \xrightarrow{p} Q^{-1}g, \quad \text{so} \quad \hat{\beta} \xrightarrow{p} \beta + Q^{-1}g,$$

which is inconsistent for β because $Q^{-1}g \neq 0$.

3. For this question, use the working approximation $se(n) = c/\sqrt{n}$. At $n = 225$, an estimator's standard error is 0.24.
- (a) What would its standard error be at $n = 900$ and at $n = 2025$?
 - (b) What would happen if the original sample size were divided by four?
 - (c) Approximately what sample size is required to reduce the standard error from 0.24 to 0.10?

Solution. Under the working approximation $se(n) = c/\sqrt{n}$, ratios of standard errors are inverse square roots of ratios of sample sizes.

- (a) Since $900 = 4(225)$, the approximate standard error is $0.24/2 = 0.12$. Since $2025 = 9(225)$, it is $0.24/3 = 0.08$.
- (b) Dividing n by four approximately doubles the standard error, from 0.24 to 0.48.
- (c) Let n_1 be the required size. Then

$$\frac{0.10}{0.24} = \sqrt{\frac{225}{n_1}}, \quad n_1 \simeq 225 \left(\frac{0.24}{0.10} \right)^2 = 225(5.76) = 1296.$$

Thus the working approximation requires about 1,296 observations.

4. Let (X_t) have

$$\mathbb{E}(X_t) = 2, \quad \text{Var}(X_t) = 9, \quad \text{Cov}(X_t, X_{t-h}) = 9(0.6)^{|h|}.$$

- (a) Find the covariance and correlation between observations five periods apart.
- (b) Explain why the stated first and second moments satisfy weak stationarity.
- (c) Write $\text{Var}(\bar{X}_n)$ in terms of the covariances at different lags. How does positive serial correlation change it relative to independent observations with variance 9? Show that it tends to zero for this process. *Hint:* there are $n - h$ pairs of observations at lag h ; each covariance appears twice in the variance of the sum. You may use $\sum_{h=1}^{\infty} r^h = r/(1 - r)$ for $0 < r < 1$.
- (d) Why does weak stationarity alone not guarantee a law of large numbers? Give a simple counterexample.

Solution.

- (a) At lag five,

$$\text{Cov}(X_t, X_{t-5}) = 9(0.6)^5 = 0.69984.$$

Since both variances are 9, the corresponding correlation is

$$\text{Corr}(X_t, X_{t-5}) = \frac{9(0.6)^5}{\sqrt{9}\sqrt{9}} = (0.6)^5 = 0.07776.$$

- (b) The mean is the constant 2, the variance is the constant 9, and the covariance between two observations depends on their dates only through the lag $|h|$. These are precisely the first- and second-moment requirements for weak stationarity.

(c) Writing $\gamma_h = 9(0.6)^{|h|}$,

$$\begin{aligned}\text{Var}(\bar{X}_n) &= \frac{1}{n^2} \left[n\gamma_0 + 2 \sum_{h=1}^{n-1} (n-h)\gamma_h \right] \\ &= \frac{9}{n^2} \left[n + 2 \sum_{h=1}^{n-1} (n-h)(0.6)^h \right].\end{aligned}$$

Every extra covariance term is positive, so for $n > 1$ this exceeds the independent-observation variance $9/n$. Moreover,

$$\text{Var}(\bar{X}_n) \leq \frac{9}{n} \left[1 + 2 \sum_{h=1}^{\infty} (0.6)^h \right] = \frac{36}{n} \rightarrow 0.$$

Thus $\bar{X}_n \xrightarrow{\text{m.s.}} 2$, and hence $\bar{X}_n \xrightarrow{P} 2$, for this particular covariance pattern.

- (d) Weak stationarity controls how moments depend on dates, but does not by itself make dependence fade with distance. For example, let $X_t = 2 + A$ for every t , where A has mean zero and positive finite variance. This process is weakly stationary, yet $\bar{X}_n = 2 + A$ for every n , so the sample mean does not converge in probability to 2.